

Trimming the Tree to Help It Grow: Selective Exit and Productivity Growth in China's State Sector *

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Abstract

Between 1998 and 2007, productivity among Chinese state-owned enterprises (SOEs) rose rapidly. Did surviving SOEs improve, or did productivity rise because unproductive firms exited? We develop a decomposition that attributes productivity growth to the primitives governing firm dynamics: within-firm productivity transitions, exit behavior, and entry composition. Applying the framework to Chinese manufacturing data, we find that selective exit is the dominant driver of state-sector productivity growth. Exit accounts for the majority of productivity gains during the reform decade, with the effect concentrated in the early years of restructuring when low-productivity firms were systematically removed. Improvements among surviving firms become important only later. Standard accounting decompositions substantially understate the role of exit because they fail to account for declining productivity trajectories prior to firm exit.

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1 Introduction

A large literature attributes China's rapid economic growth to the reallocation of resources from inefficient state-owned enterprises (SOEs) toward more productive private firms. In this view, state firms survived despite low productivity because they benefited from preferential access to credit, subsidies, and regulatory protection,¹ generating substantial misallocation of resources across firms and sectors (Hsieh and Klenow, 2009; Song et al., 2011). As market-oriented reforms gradually reduced these distortions, resources flowed toward more productive private firms, raising aggregate productivity.

Yet the evolution of productivity within the state sector itself presents a striking puzzle. During the major SOE reform period from 1998 to 2007, the productivity of state-owned firms converged substantially toward that of private firms, even as the state sector's share of output declined sharply (Hsieh and Song, 2015). If SOEs were fundamentally less efficient due to soft budget constraints and political objectives, why did their productivity improve so dramatically over time?

Understanding this convergence requires distinguishing between two competing mechanisms. The first is *selection*. If poorly performing state firms exit the market, the average productivity of the remaining firms rises mechanically even if no individual firm becomes more productive. The second is *within-firm improvement*: surviving state firms may genuinely become more efficient due to governance reforms, corporatization, or stronger market discipline. These mechanisms have very different economic interpretations. If convergence reflects selection, productivity growth arises primarily from the removal of inefficient firms. If instead surviving firms improved their performance, then reforms may have fundamentally altered the efficiency of state ownership itself.

Several features of the data suggest that selection may play an important role. Between 1998 and 2007, more than three quarters of state-owned firms exited from the industrial census data, reflecting one of the largest episodes of industrial restructuring in modern economic history. Directly studying exit rates over time reveals that during the reform

¹It is well documented in the literature that state-owned firms had preferential access to inputs such as labor, capital, land and export licenses. See Bai et al. (2006), Hsieh and Klenow (2009), Song et al. (2011), Brandt and Zhu (2001), and Khandelwal et al. (2013) for details.

period, exit rates were high and selective. Exit rates peaked at over 40% among the least productive firms at the start of the reform, and were highly selective, dropping to under 15% for the most productive firms. The intensity and degree of selection declined by the end of the reform: exit rates were around 30% among the least productive, and 10% among the most productive. However, even this degree of selection could still be sufficient to play a meaningful role in aggregate productivity growth.

At the same time, the productivity trajectories of surviving firms display a different pattern. Conditional on survival, the productivity of state-owned firms rises steadily throughout the reform period. This upward trend suggests that some firms may have genuinely improved their performance, potentially reflecting governance reforms such as the establishment of the State-Owned Assets Supervision and Administration Commission (SASAC) in 2003 and subsequent corporate restructuring policies. Taken together, the data suggest that both selection and within-firm improvement may contribute to the observed productivity convergence.

Quantifying the relative importance of these mechanisms is challenging. A large literature has developed accounting decompositions of aggregate productivity growth that attribute changes to within-firm improvements, reallocation, entry, and exit ([Baily et al., 1992](#); [Griliches and Regev, 1995](#); [Foster et al., 2001](#); [Melitz and Polanec, 2015](#)). Applying these methods to our setting yields ambiguous conclusions. When we decompose productivity changes over the full reform period, exit appears to play only a modest role. However, when the same decomposition is applied to shorter time intervals, the contribution of exit increases substantially. The resulting conclusions depend sensitively on the choice of time horizon, making it difficult to draw clear economic interpretations about the mechanisms driving productivity growth.

This ambiguity arises because of two limitations faced by traditional accounting decompositions. First, as selection becomes stronger, these decompositions tend to reassign the effect of exit to between-firm reallocation, obscuring the role of market selection. Second, over long horizons they mechanically drive the contribution of exit toward zero, even when selective exit is central to aggregate productivity growth.

This paper develops a decomposition framework designed to address these limitations. Our approach focuses on the primitives governing the evolution of the productivity distribu-

tion: the productivity transition process $P(z'|z)$, the exit rule $Q(z)$, and the entry process $\lambda G(z)$. These objects correspond to the core mechanisms governing firm dynamics in a broad class of heterogeneous-firm models, including the canonical framework of [Hopenhayn \(1992\)](#). Rather than deriving them from a specific structural model, we treat these primitives as reduced-form objects that summarize the evolution of the productivity distribution in the data. By constructing modular counterfactuals that vary one primitive at a time while holding the others fixed, the decomposition isolates the contribution of each mechanism to aggregate productivity growth.

We then show how these primitives can be identified and estimated nonparametrically from firm-level panel data. The key insight is that the productivity transition process and exit rule jointly determine the observed evolution of firm productivity among surviving firms and the observed exit probabilities. Under mild conditions, these relationships can be inverted to recover the underlying primitives. Once these primitives are estimated, the decomposition can be implemented through a sequence of counterfactual exercises that isolate the contribution of productivity transitions, exit behavior, and entry composition.

We apply the framework to firm-level data from China’s Annual Survey of Industrial Firms during the major SOE reform period from 1998 to 2007. Our results reveal that selective exit is the dominant driver of productivity growth within the state sector over the full reform period. We estimate that approximately 85 percent of the observed productivity increase can be attributed to changes in the exit margin, while 15 percent reflects within-firm productivity improvements and close to zero percent reflects entry.

The results also reveal a clear sequencing pattern over time. During the early reform years (1998–2002), productivity growth is driven primarily by intensified selection through firm exit. In later years (2003–2007), productivity improvements among surviving firms becomes the dominant drive of growth.

To move from a statistical decomposition to a deeper economic interpretation, we use a Hopenhayn-style model to connect institutional reforms to the forces governing firm dynamics. Interpreted through this lens, our results align closely with the timing and objectives of the major SOE reforms of this period. The early phase of the reform (1998–2002) is consistent with the “grasp the large, let go of the small” policy, which promoted the exit of small, un-

productive SOEs through harder budget constraints and stricter bankruptcy enforcement, while also encouraging consolidation among larger and more productive firms. The later phase (2003–2007) is instead consistent with reforms designed to improve corporate governance, including the creation of the State-Owned Assets Supervision and Administration Commission (SASAC) in 2003 and the split-share structure reform in 2005.

These findings suggest a reinterpretation of China’s state-sector transformation. Rather than reflecting immediate efficiency improvements within state firms, the early productivity gains of the reform period primarily arose from the removal of poorly performing firms. Only after this initial selection phase do within-firm improvements become an important driver of productivity growth. In this sense, China’s reform experience resembles a process of “trimming the tree to help it grow”: market selection first removes unproductive firms, creating conditions under which surviving firms can subsequently improve their performance.

Beyond the Chinese context, the framework developed in this paper provides a general tool for analyzing productivity dynamics in environments where policy reforms alter the mechanisms governing firm entry, exit, and productivity evolution. By focusing on primitive-level changes rather than realized productivity flows, the decomposition provides a bridge between reduced-form empirical analysis and structural models of firm dynamics.

The remainder of the paper proceeds as follows. Section 1.1 goes through the related literature. Section 2 provides institutional background on China’s state-owned enterprise reforms. Section 3 describes the data and documents the key empirical patterns that motivate our analysis. Section 4 introduces our decomposition framework. Section 5 describes identification and estimation of the framework primitives. Section 6 presents the decomposition results, and compares them against what standard accounting decompositions show. Section 7 interprets the findings through the lens of a Hopenhayn model. Section 8 concludes.

1.1 Related Literature

Our paper contributes to two related strands of literature: the sources of China’s reform-era growth and the decomposition of aggregate productivity change. Empirically, we document convergence in total factor productivity (TFP) between the state and private sectors

from 1998 to 2007, with the state sector growing faster over this period. Methodologically, we develop a semi-structural decomposition that identifies the mechanisms behind that convergence.

A large literature attributes China's growth during the reform era to marketization and the expansion of private economic activity. At the micro level, studies emphasize the role of reallocation across firms. [Hsieh and Klenow \(2009\)](#) highlight the productivity costs of distortions that prevent resources from flowing to more productive producers, while [Brandt et al. \(2012\)](#) document substantial productivity gains from firm entry and reallocation in Chinese manufacturing. At a more aggregate level, transition models such as [Song et al. \(2011\)](#) interpret China's growth as driven by the rise of high-productivity entrepreneurial firms alongside a gradually contracting state sector. Related accounts emphasize the expansion of private business and the institutional reforms that enabled it ([Lardy, 2014](#); [Xu, 2011](#)). In this view, China's growth is primarily a story of private-sector expansion and market-driven reallocation.

A second strand emphasizes instead the transformation of the state sector itself. [Hsieh and Song \(2015\)](#) show substantial convergence in productivity between surviving state firms and private firms over 1998–2007, during a period of major SOE restructuring. More broadly, [Naughton and Tsai, eds \(2015\)](#) and [Lin et al. \(2020\)](#) argue that persistent state ownership remained a central feature of China's development model. Our paper contributes to this literature by moving from documenting convergence to identifying its sources. We show that state-sector TFP convergence was driven not by a single mechanism, but by a sequence of mechanisms: exit dominates over the reform era as a whole, while a finer decomposition reveals that early gains were driven mainly by exit, consistent with harder budget constraints, and later gains increasingly by within-firm improvement, consistent with corporatization, governance reform, and stronger managerial incentives. In this sense, our results reconcile two views of SOE reform: one centered on selection through discipline ([Maskin and Xu, 2001](#); [Kornai et al., 2003](#)), and the other on productivity gains within surviving firms ([Groves et al., 1994](#); [Bai et al., 2004](#); [Hsieh and Song, 2015](#)).

Our second contribution is methodological. The classic productivity-decomposition literature partitions aggregate productivity growth into within-firm improvement and realloca-

tion, often augmented with entry and exit terms. Foundational contributions include [Bailey et al. \(1992\)](#), [Griliches and Regev \(1995\)](#), and [Foster et al. \(2001\)](#). [Melitz and Polanec \(2015\)](#) further show that standard dynamic decompositions can misstate the contributions of entry and exit because they benchmark turnover against a common reference level rather than contemporaneous surviving firms, a distortion that worsens over longer horizons. More broadly, existing accounting decompositions are transparent but descriptive: they decompose realized productivity change rather than the effects of changes in the underlying mechanisms governing firm dynamics.

That limitation is particularly important in our setting. China’s state-sector reform involved large changes in ownership, survival, and firm composition, so standard accounting decompositions are especially vulnerable to two problems. First, their estimates can be sensitive to the horizon over which firms are classified as entrants or exiters, since longer windows mechanically pool heterogeneous cohorts. Second, because incumbents, entrants, and exiters are themselves shaped by reform, decompositions based on realized cohorts can confound changes in underlying primitives with changes in the set of firms being compared. We show that these issues are quantitatively important in our context and can obscure the sources of growth within the state sector.

Our approach is therefore best understood as semi-structural. Like the accounting literature, it remains close to observed firm-level data. But like model-based approaches ([Basu and Fernald, 2002](#); [Petrin and Levinsohn, 2012](#); [Baqaee and Farhi, 2020](#)), it evaluates counterfactual trajectories generated by economically meaningful primitives. Rather than asking only how realized productivity growth can be partitioned into within, between, entry, and exit components, we ask how aggregate productivity would have evolved under alternative paths for within-firm productivity transitions, exit behavior, and entry. This yields a decomposition that remains transparent and data-driven while delivering a clearer account of China’s SOE reform era: over the full decade, exit is the main driver of state-sector productivity growth, but over shorter horizons the dominant mechanism shifts from selection-driven gains to within-firm improvement.

2 Institutional Background: State-Owned Enterprise Reform, 1998–2007

The reform of state-owned enterprises (SOEs) in China began in 1978. Prior to reform, the urban industrial economy operated under a centrally planned system in which production, input allocation, prices, and employment were heavily controlled by the state. SOEs faced weak competitive pressure, managers had limited autonomy, and incentives were often poorly aligned with profitability or productivity. Because firms were expected to fulfill administrative objectives and losses were routinely accommodated through fiscal support or directed bank lending, budget constraints were soft and inefficient firms could continue operating without meaningful risk of exit ([Lau et al. \(2000\)](#); [Brandt and Zhu \(2001\)](#); [Naughton \(2007\)](#)).

Early reforms targeted efficiency gains and a gradual movement toward more market-oriented behavior for firms. At the same time, they also reflected a broader strategy of “reform without losers,” motivated by concerns about social stability. From 1978 to 1993, reforms were introduced that expanded enterprise autonomy, decentralized administrative control to local governments, and implemented managerial responsibility systems that partially linked compensation for both managers and workers to firm performance, all while constraining deviations from the planned-economy framework. Concurrently, the introduction of a dual-track pricing system – under which firms operated partly at market prices and partly under planned allocations – exposed SOEs to market incentives while deliberately limiting the speed and scope of adjustment. Under this gradualist approach, loss-making SOEs were typically supported through directed credit from state-owned banks and implicit government guarantees rather than being allowed to fail or shed labor ([Lau et al. \(2000\)](#); [Brandt and Zhu \(2001\)](#)). As a result, while early reforms encouraged efficiency gains and nudged firms toward market-oriented behavior, budget constraints remained soft and exit was largely absent: [Hsieh and Song \(2015\)](#) report that the annual exit rate among state-owned firms was only around 0.9% between 1991 and 1995. This prevented market selection from playing a central role in reallocating resources within the state sector ([Groves et al.](#)

(1994); [Jefferson et al. \(1996\)](#); [Li \(1997\)](#)). At the same time, this incremental transition moved the economy away from a fully planned system toward a hybrid regime combining administrative control with market mechanisms, shaping the institutional environment in which subsequent, more far-reaching reforms would take place.

By the mid-1990s, this gradual reform strategy had become increasingly difficult to sustain. Many SOEs remained chronically unprofitable and imposed growing fiscal burdens on local governments and the banking system ([Berkowitz et al. \(2017\)](#)). Continued protection of underperforming firms became increasingly costly, setting the stage for a more radical restructuring of the state sector in the late 1990s ([Lardy \(2014\)](#)).

Selection and Restructuring

The centerpiece of the late-1990s restructuring was the policy known as “grasp the large, let go of the small” (*zhuā dà fàng xiǎo*), formally introduced at the 1999 Fourth Plenum of the Communist Party’s Central Committee. The policy differentiated SOEs by size and strategic importance. Large SOEs in key sectors were consolidated into enterprise groups and reorganized through corporatization, while small and medium-sized SOEs were privatized, merged, or closed ([Naughton \(2007\)](#); [Hsieh and Song \(2015\)](#)). This “grasp the large” approach involved not only restructuring existing firms into limited liability or joint-stock companies but also establishing new large state-owned enterprises in strategic sectors ([Hsieh and Song \(2015\)](#), [Rubens \(2023\)](#)). Meanwhile, local governments were granted expanded authority to restructure smaller firms, and many loss-making enterprises were liquidated or transferred to private ownership.²

These restructuring efforts were accompanied by broader institutional changes that made firm exit more feasible. Beginning in the late 1990s, the central government implemented enterprise restructuring programs that expanded the use of bankruptcy procedures and supported large-scale layoffs through reemployment programs and an emerging social safety net. Millions of workers were displaced from the state sector, and the number of state-

²[Gu and Jia \(2022\)](#) provide a structural analysis of the corporatization and privatization margins, showing that large SOEs could rationally prefer corporatization to preserve borrowing capacity, helping explain how the “grasp the large, let go of the small” strategy could be seen as consistent with endogenous profit maximizing behavior.

owned firms declined sharply during this period ([Hsieh and Song \(2015\)](#); [Brandt and Rawski \(2008\)](#)). Together, these reforms hardened budget constraints and introduced a meaningful exit margin for poorly performing firms, allowing market selection to play a substantially larger role in reallocating resources within the state sector.

Governance and Ownership Reforms

Following the initial wave of restructuring, a second phase of reforms in the early 2000s focused on improving the governance and performance of the remaining state enterprises. A major institutional development was the establishment of the State-Owned Assets Supervision and Administration Commission (SASAC) in 2003. SASAC consolidated previously fragmented state ownership, which had been dispersed across multiple government ministries and local bureaucracies, and assumed responsibility for managing and supervising central SOEs as the formal representative of the state as owner ([Naughton \(2006\)](#); [Lardy \(2019\)](#)). SASAC enhanced oversight of state assets and implemented corporate governance reforms, including restructuring boards and introducing new performance evaluation systems for SOE managers.

During this period, many SOEs were reorganized as limited liability or joint-stock companies under the framework of China's Company Law, and corporate governance structures were gradually formalized through the introduction of boards of directors and clearer separation between ownership and management ([Naughton \(2007\)](#)). Complementary capital market reforms further strengthened governance. In particular, regulations introduced requirements for independent directors on the boards of listed companies, bringing greater external oversight and professional management practices to SOEs ([Bhabra and Li \(2011\)](#)).

This was followed in 2005 by the split-share structure reform, which addressed a fundamental distortion in the governance of publicly listed SOEs. Prior to the reform, listed SOEs operated under a dual-share structure in which roughly two-thirds of shares were held by the state as non-tradable blocks, while only the remaining one-third were freely traded by private investors. Because managerial evaluation and promotion were formally linked to overall firm value, and non-tradable state shares were effectively valued on the basis of book

asset values rather than market prices, the link between market valuation and managerial incentives was weak. This dual structure created an incentive misalignment, systematically rewarding asset accumulation over productive efficiency.³ The split-share structure reform dismantled this dual system by converting non-tradable state shares into tradable shares. By exposing state-held shares to market pricing, the reform strengthened the link between managerial incentives and market-based performance. In doing so, it shifted incentives away from asset expansion and toward productivity and profitability.

Taken together, these reforms fundamentally reshaped China's state sector during the late 1990s and early 2000s, leading to a substantial reduction in the number of state-owned firms and significant changes in the governance and organization of those that remained.

3 Data, Measurement, and Empirical Patterns

This section describes the data, measurement, and sample construction used in the analysis, and then documents the main empirical patterns that motivate our framework. We begin by introducing the firm-level data, ownership classification, and productivity measures. We then show that state-owned enterprises converged substantially toward private firms in productivity between 1998 and 2007, and present descriptive evidence on the margins through which this convergence may have occurred.

3.1 Data, Measurement, and Sample Construction

Our analysis uses the Annual Survey of Industrial Firms (ASIF) from China for the years 1998–2007. The ASIF is an annual survey of manufacturing enterprises conducted by the National Bureau of Statistics of China and is one of the most important sources of information

³The structure also generated broader market distortions including ineffective boards, profit tunneling through related-party transactions, rampant stock market speculation, blocked merger activities, and stunted development of corporate bond and derivative markets (Liao et al. (2014)).

on the performance of China's manufacturing sector.⁴ The data provide annual firm-level observations for all state-owned enterprises and for private firms exceeding 5 million RMB in annual revenue, allowing systematic comparison of state and private firm performance. This includes a large and expanding set of manufacturing firms, with coverage increasing from roughly 100,000 to over 300,000 firms over the sample period. In addition to substantial firm entry, a key feature of the data is the prevalence of exit: among state-owned firms observed in 1998, approximately 80% exit by 2007.

The unit of observation in the data is a registered firm in a given year. Within state-owned industrial groups, each firm owned by the group is a separate observation in our data. Our baseline variables of interest include the firm's industry at the four-digit level,⁵ location, age, ownership status, total assets, total liabilities, industrial production, value added, revenue, number of employees, wage payments, capital stock, intermediate goods purchase, export revenues, and subsidies.⁶ After cleaning the data to remove observations with missing or implausible values for key production variables, our final sample contains over 1.7 million firm-year observations.

Our data treatment follows [Hsieh and Klenow \(2009\)](#) (henceforth HK). For the baseline analysis, we define capital stock as the book value of fixed capital net of depreciation. As an alternative, we employ the perpetual inventory method with an annual depreciation rate of 9% and province-sector-specific investment rates (see the online appendix from [Brandt et al. \(2012\)](#)).

For labor input, we adjust the reported wage payments to account for non-wage benefits, scaling up wages to match the 50% aggregate labor share from Chinese input-output tables. As a robustness check, we also measure labor input using employment.

Our analysis requires precise identification of firm ownership. We adopt the ownership classification from [Hsieh and Song \(2015\)](#), which provides a consistent framework for distinguishing state-owned and privately-owned firms (see Appendix B.1 for a detailed discussion

⁴We construct the panel using a method that closely follows the approach outlined in [Brandt et al. \(2012\)](#).

⁵To preserve greater sectoral granularity in our analysis, we follow [Liu \(2019\)](#) and map the original 4-digit industry codes to an expanded set of 80 2-digit codes, rather than using the standard classification of 30 2-digit sectors in manufacturing.

⁶Data on value added and industrial production in 2004 are imputed since they are missing in the original data.

of the complexities in defining state ownership in China, and note that throughout the paper, we use the terms “private firms,” “privately-owned firms,” and “non-state-owned firms” interchangeably). To isolate ownership effects from transition effects, we restrict attention to firms that maintained consistent ownership throughout the sample period – either “forever state-owned” or “forever private-owned” – and exclude firms that changed ownership status during the transition.⁷

For our baseline analysis, we measure firm-level productivity following HK. However, recognizing that their assumptions of constant returns to scale and homogeneous factor shares within sectors may be restrictive, we also employ an alternative approach. Following [Brandt et al. \(2017\)](#), we estimate production functions that allow for varying returns to scale and heterogeneous factor shares, deriving firm-level revenue TFP (TFPR) as a residual (see Appendix B.2). This alternative specification serves as a robustness check for our main results.

3.2 Empirical Patterns

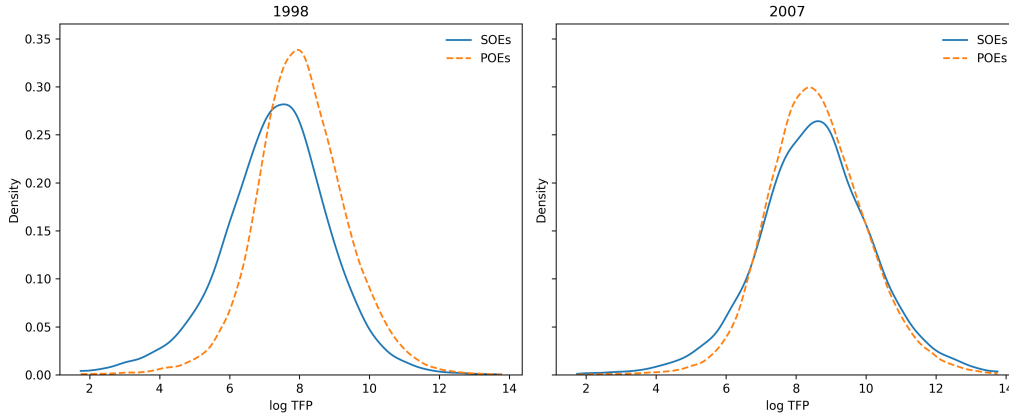
This subsection documents the empirical patterns that motivate our analysis. We first show that between 1998 and 2007, the productivity distribution of state-owned enterprises (SOEs) converged substantially toward that of privately owned enterprises (POEs), and that this convergence occurred broadly across industries.⁸ We then present descriptive evidence on the margins through which this catch-up may have occurred, focusing on within-firm productivity growth, selective exit, and entry. These patterns motivate the decomposition framework developed in the rest of the paper.

⁷While privatization played an important role in China’s aggregate productivity growth during this period (see [Chen et al., 2021](#); [Wang et al., 2024](#)), our focus here is instead on comparing productivity dynamics among firms with persistent ownership status, in order to assess how state-owned enterprises evolved relative to a private-sector benchmark in the presence of state-sector-specific reforms.

⁸[Hsieh and Song \(2015\)](#), [Chen et al. \(2021\)](#), [Brandt et al. \(2022\)](#) and [Gu and Jia \(2022\)](#) have similar findings that state-owned and privatized firms started from significantly lower TFP levels but then caught up with always-private firms by 2007. [Brandt et al. \(2022\)](#) find that the TFP gap between state-owned and private manufacturing firms narrowed and even became slightly positive after 2007. However, [Chen et al. \(2021\)](#) and [Brandt et al. \(2022\)](#) interpret the convergence in TFP across ownership types with caution. See Appendix C for detailed discussion.

3.2.1 Convergence Between SOEs and POEs

Figure 1: TFP Convergence Between State-Owned and Private Firms, Economy-Wide



We begin by documenting convergence in productivity between state-owned enterprises (SOEs) and privately-owned enterprises (POEs) between 1998 and 2007.

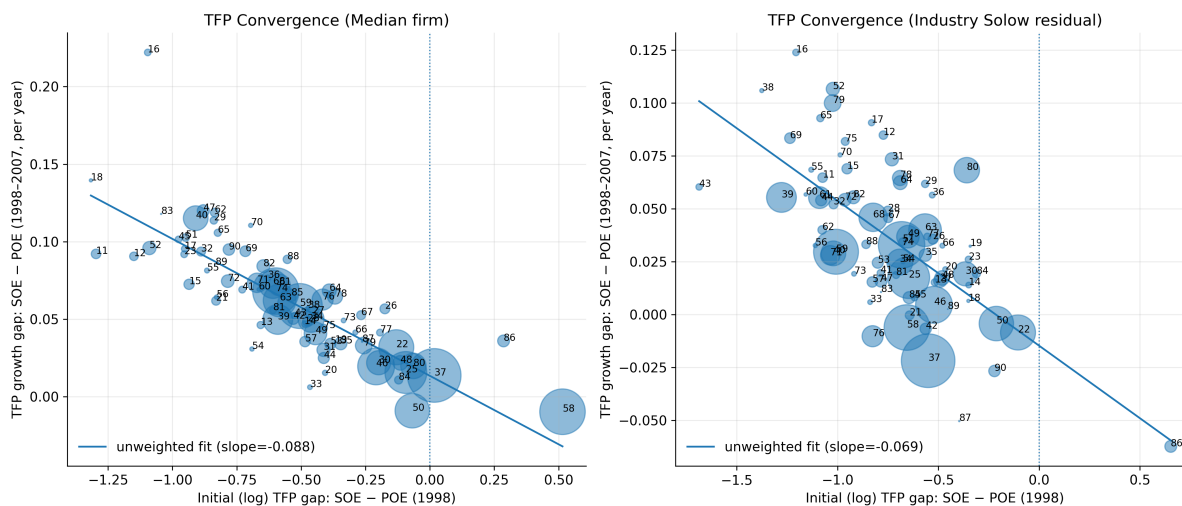
Figure 1 compares the distributions of log TFP for SOEs and POEs in 1998 and 2007. In 1998, SOEs lag substantially behind POEs: the mean log TFP of SOEs is 7.284, compared with 8.047 for POEs. Over the subsequent decade, both groups experience productivity growth, but SOEs improve at a faster rate. By 2007, the two distributions nearly coincide. The mean log TFP of SOEs rises to 8.533, while the mean for POEs rises to 8.591. Thus, the SOE-POE mean gap falls from 0.763 log points in 1998 to just 0.058 log points in 2007. While the overlap is not exact – the SOE distribution remains somewhat more dispersed – the two groups are nearly identical in mean productivity by the end of the sample.

To assess whether convergence occurred broadly across industries, Figure 2 plots sector-level convergence patterns.⁹ For each sector, we compute the initial gap in productivity between SOEs and POEs in 1998 and the subsequent annualized change in that gap between 1998 and 2007. The figure shows a strong negative relationship: sectors in which SOEs initially lagged further behind experienced faster subsequent catch-up. This pattern holds whether productivity is measured using sector-level median TFP or an aggregated industry-level Solow residual.

⁹Appendix ?? documents distributional convergence patterns separately for each manufacturing industries.

Taken together, these facts show that SOEs began the reform period substantially less productive than POEs but largely closed the gap by 2007, and that this convergence occurred across most industries rather than being driven by a small subset of sectors.

Figure 2: TFP Convergence Between State-Owned and Private Firms, Across Sectors



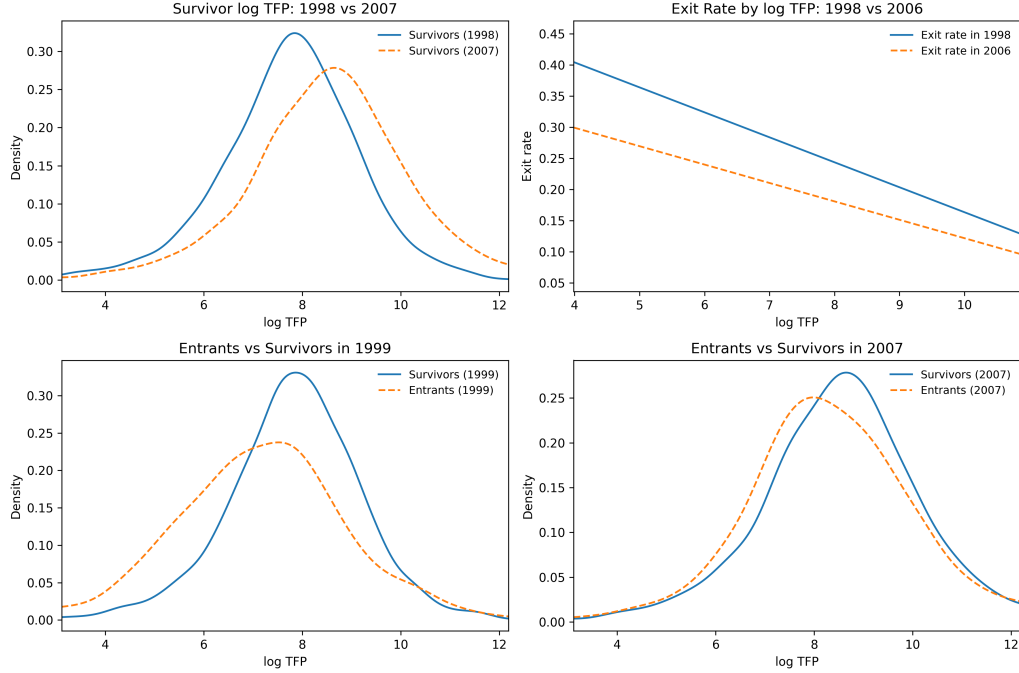
Notes: The size of each marker represents the market share of the sector in 1998. Median gives the median productivity of firms in the sector. We exclude tobacco manufacturing (24) since it is a state-controlled sector. The unweighted line gives an unweighted OLS fit through the data. The weighted line weights by the market share of the sector in 1998.

3.2.2 Stylized Facts on the Sources of Convergence

We next present descriptive evidence on the margins through which this convergence may have occurred. Figure 3 summarizes three key patterns.

First, surviving SOEs (those operating in both 1998 and 2007) experienced substantial productivity growth. The top-left panel compares the distribution of log TFP among SOEs that are present in both 1998 and 2007. This distribution shifts markedly to the right over time. The mean log TFP of surviving SOEs rises from 7.716 in 1998 to 8.499 in 2007, an increase of 0.783 log points. This suggests that at least part of the observed convergence reflects genuine productivity gains among continuing firms rather than purely compositional change.

Figure 3: TFP Convergence Between State-Owned and Private Firms



Second, exit patterns suggest an important role for selection. The top-right panel plots a linear approximation to the exit rule as a function of TFP for the years 1998 and 2006. Exit rates are high and strongly declining in productivity in both periods, indicating substantial selection on performance. In 1998, the fitted exit rule has intercept 0.565 and slope -0.040, while by 2007 the intercept falls to 0.417 and the slope becomes flatter at -0.030. Thus, exit is both more prevalent and more strongly tilted toward low-productivity firms early in the reform period. This pattern is consistent with a stronger role for selective exit in the earlier years of reform.

Third, entry appears less likely to account for the catch-up of SOEs to POEs. The bottom panels compare entrants with surviving incumbents for the years 1999 and 2007. In 1999, the mean log TFP of entrants is 7.097, well below the 1998 mean of 7.716 for surviving SOEs. By 2007, entrants are closer to survivors, but still somewhat less productive: mean log TFP is 8.346 for entrants, compared with 8.499 for survivors. These patterns suggest that entry did not drive convergence and may, if anything, have worked against it.

Taken together, these facts indicate that both within-firm productivity growth and selective exit plausibly contributed to the convergence between SOEs and POEs, while entry

appears to have potentially worked against convergence.

While these stylized facts suggest several candidate mechanisms, they do not reveal their relative quantitative importance. A natural next step is therefore to apply an existing accounting decomposition to the data. As we show in the next section, however, standard accounting decompositions can be misleading in our setting, because strong selection over long horizons can cause them to understate the role of exit in productivity growth. This motivates the alternative framework we develop later in the paper.

4 Decomposition Framework

This section develops a minimal framework to describe how the distribution of firm-level productivity evolves and how aggregate changes can be attributed to distinct underlying mechanisms. Our goal is diagnostic rather than structural. We aim to summarize how much of the observed evolution in productivity can be traced to each of three economic primitives – within-firm productivity improvement, firm exit, and firm entry – and to provide a way to interpret these components when thinking about policy.

The framework has two ingredients. First, a statistical environment that summarizes how the productivity distribution changes over time. Second, a decomposition based on modular counterfactuals that isolates the contribution of each primitive while keeping the others fixed.

To demonstrate the generality and flexibility of the statistical environment, we show how it is sufficient to fully characterize the dynamics of the productivity distribution in a variation of the canonical heterogeneous-firm model of industry dynamics developed by [Hopenhayn \(1992\)](#).

We conclude the section by going through two examples that demonstrate how existing accounting decompositions can mislead in settings where selective exit plays a dominant role in growth.

4.1 The Primitives, Timing, and Law of Motion

4.1.1 The Primitives

We model the evolution of the productivity distribution $\pi(z)$ through three primitives: the productivity transition process, the exit rule, and the entry process.

The productivity transition process, $P(z'|z)$: the probability that a firm with productivity z today has productivity z' tomorrow. This kernel summarizes all forces affecting within-firm productivity evolution, such as management quality, technology adoption, and learning.

The exit rule, $Q(z)$: the probability that a firm with productivity z leaves the market. This nests deterministic exit for firms with productivity below some fixed threshold $\underline{z}$ as a special case. Allowing for probabilistic exit enables our framework to match the widely observed empirical pattern that firms with similar productivities often make different exit decisions, and accommodates, for instance, heterogeneity in local conditions, exposure to government safety nets, or expectations about future profitability.

The entry process, $\lambda G(z)$: the mass and productivity distribution of new entrants. The scalar λ determines the mass of entrants relative to incumbents, while $G(z)$ captures the productivity distribution of entrants, reflecting the capabilities available to new firms.

4.1.2 Timing

In the literature there are two common conventions for the timing of exit decisions. In models such as [Hopenhayn \(1992\)](#) and [Ericson and Pakes \(1995\)](#), firms decide whether to exit based on the expected continuation value before the next productivity shock is realized. In other models, firms first observe the new productivity realization and then decide whether to continue or exit ([Jovanovic, 1982](#); [Clementi and Palazzo, 2016](#); [Asturias et al., 2023](#); [Gu and Jia, 2022](#)). We adopt the latter timing as our benchmark specification, which implies that exit decisions are made in response to productivity shocks that are unobserved to the

econometrician.¹⁰

In our language, incumbent firms first experience productivity shocks according to $P(z'|z)$, then decide whether to continue or exit based on the new draw z' and exit rule $Q(z')$. Finally, new firms enter with productivity z drawn from distribution $G(z)$. Placing entry last simplifies the framework by abstracting from two complications: (i) entrants that might immediately exit, and (ii) entrants whose productivity could change before they are first observed. This timing handles (i) by treating such firms as never having entered, and handles (ii) by interpreting entrants' first observed productivity as their productivity upon entry.

4.1.3 Evolution of the Productivity Distribution

Under our baseline timing, the evolution of the productivity distribution can be expressed as a simple updating rule that serves as the foundation of our proposed decomposition. Define the unnormalized mass of firms with productivity z in the next period as

$$M(z|\pi, P, Q, \lambda G) = \lambda G(z) + \int \pi(z')P(z|z')(1 - Q(z))dz'. \quad (1)$$

The first term captures the inflow of new entrants with productivity z ; the second term aggregates surviving incumbents that transition to (or remain at) productivity z . The updated productivity distribution is then

$$\pi'(z|\pi, P, Q, \lambda G) = \frac{M(z|\pi, P, Q, \lambda G)}{\int M(z'|\pi, P, Q, \lambda G)dz'}. \quad (2)$$

Equation (2) summarizes how the primitives $(P, Q, \lambda G)$ map into the evolution of the productivity distribution. This serves as the backbone of all counterfactual and decomposition exercises that follow: by varying one primitive at a time while holding the others fixed, we can trace how each mechanism – within-firm transitions P , exit behavior Q , and entry λG – contributes to the observed evolution of productivity.

¹⁰In future drafts, we plan to explore a continuous-time formulation in which shocks and exit decisions occur in continuous time, which removes the need to specify a discrete timing convention.

4.1.4 Hopenhayn in Our Framework

To demonstrate the generality and flexibility of our framework, we now show as an example how it is sufficient to fully characterize the dynamics of the productivity distribution in the canonical [Hopenhayn \(1992\)](#) environment under our preferred timing.

Consider a competitive industry populated by firms that differ in productivity $z \in \mathcal{Z}$. Let p_t denote the output price and w_t denote the input price in period t . A firm with productivity z earns current profits

$$\tilde{\pi}(z; p_t, w_t) - f_t, \quad (3)$$

where f_t is a fixed operating cost. Firms discount future payoffs at rate $\beta \in (0, 1)$.

Productivity evolves stochastically over time according to a Markov process with transition kernel $\tilde{P}_t(z'|z)$. Given prices, the value of an incumbent firm satisfies

$$V_t(z) = \tilde{\pi}(z; p_t, w_t) - f_t + \beta \int \max\{V_{t+1}(z'), 0\} \tilde{P}_t(dz'|z). \quad (4)$$

A firm exits whenever the continuation value is negative. The equilibrium survival rule can therefore be written as

$$\chi_t(z') = \mathbf{1}\{V_{t+1}(z') \geq 0\}. \quad (5)$$

Potential entrants pay an entry cost k_t and draw productivity from distribution $\tilde{G}_t$. Since entrants pay the cost before observing productivity, the free-entry condition states

$$k_t = \int V_t(z) \tilde{G}_t(z) dz. \quad (6)$$

Equilibrium also requires factor and output markets to clear, so prices, firm values, entry and exit are jointly determined.

Although equilibrium depends on many structural parameters – for example, technology, operating costs, entry costs, and discount rate – the evolution of the cross-sectional productivity distribution can be fully characterized by the primitives defined at the start of this

section.

Formally, among incumbent firms, the transition process from Hopenhayn, $\tilde{P}_t(z'|z)$, corresponds exactly to $P(z'|z)$ within our framework. The equilibrium exit rule, $1 - \chi_t(z')$, corresponds to $Q(z')$ within our framework. Finally, the entry distribution $\tilde{G}_t(z)$ corresponds to $G(z)$ within our framework, while the mass of entrants relative to incumbents that is consistent with equilibrium in the Hopenhayn model corresponds to λ within our framework. Together, these fully characterize the productivity distribution dynamics within the Hopenhayn environment.

4.2 A Modular Decomposition

For the purpose of studying the drivers of productivity growth, the framework is consistent with a number of decompositions. Consider two regimes, A and B , characterized by primitives $(P^A, Q^A, \lambda^A G^A)$ and $(P^B, Q^B, \lambda^B G^B)$. Starting from an initial distribution π_0 , compute counterfactual distributions by allowing one primitive at a time to change while holding the others fixed, and updating the distribution of productivities according to Equation (2). As an example:

Within-all effect: evolve the economy with P^B but keep Q^A and $\lambda^A G^A$ fixed.

Exit effect: then update Q from Q^A to Q^B while holding P^B and $\lambda^A G^A$ constant.

Entry effect: finally allow λG to shift from $\lambda^A G^A$ to $\lambda^B G^B$.

The difference between each step gives the contribution of the corresponding channel. Summing the three components exactly reproduces the change in the productivity distribution from switching from regime A to B:

$$\Delta\pi \equiv \pi^B - \pi^A = (within) + (exit) + (entry)$$

Because the evolution of the distribution is non-linear, the quantitative contributions depend on the order in which primitives are updated. For example, switching to the alter-

native timing of exit, within, and entry generally produces a different decomposition. There are several reasonable ways to address this issue. One approach is to report results for each ordering. Another is to use a *Shapley decomposition*, which assigns to each component its average marginal contribution across all possible orderings of the primitive changes (Shapley et al., 1953; Shorrocks et al., 2013). With three primitives, this corresponds to averaging the contributions across the six possible sequences of updates. A third approach is that, in certain contexts, institutional or economic considerations may suggest a natural ordering, in which case the corresponding decomposition can be informative on its own.

Ultimately, this procedure asks a simple counterfactual question: if only one aspect of the productivity process had changed while the others remained fixed, how much aggregate productivity change would we observe? These modular counterfactuals provide a transparent way to understand which underlying mechanisms account for the aggregate changes observed in the data, in the spirit of productivity decompositions commonly used in the firm dynamics literature (Foster et al., 2001).

4.3 Comparison to Existing Accounting Decompositions

We present two simple examples showing how standard accounting decompositions – such as Baily et al. (1992) (BHC), Foster et al. (2001) (FHK), Griliches and Regev (1995) (GR), and Melitz and Polanec (2015) (DOPD) – can give misleading answers relative to our decomposition based on the productivity transition matrix P , the exit rule Q , and the entry process λG . The first example shows that as selection becomes stronger, these decompositions tend to reassign the effect of exit to between-firm reallocation, obscuring the role of market selection. The second example shows that over long horizons, they mechanically drive the contribution of exit toward zero, even when selective exit is central to aggregate productivity growth.

In both examples, the initial cross section consists of n firms with equal mass $1/n$, and the object of interest is average productivity among active firms at horizon h .

Throughout, the pre-reform regime is

$$(P^A, Q^A, \lambda^A G^A) = (I, 0, 0),$$

so productivities are constant, there is no exit, and there is no entry. The post-reform regime $(P^B, Q^B, \lambda^B G^B)$ is example-specific. In both examples, $\lambda^B = 0$, so entry is absent and only the within component (P) and exit component (Q) are active.

To simplify exposition, we use FHK as the reference accounting decomposition.

Our decomposition uses Shapley values across orderings of $(P, Q, \lambda G)$. Since entry is absent, only two effective orderings remain: changing P first and then Q , or changing Q first and then P .

4.3.1 Example 1: Selection with No Within Growth

The purpose of this example is to show that standard accounting decompositions can mechanically relabel a pure selection event as between-firm reallocation, whereas our decomposition attributes the entire effect to the change in the exit rule. In this example, all but the most productive firm exit. Initial productivities are

$$\delta, 2\delta, \dots, n\delta.$$

Hence, aggregate productivity is

$$\Phi_0 = \frac{1}{n} \sum_{i=1}^n i\delta = \frac{n+1}{2}\delta.$$

In the post-reform regime,

$$P^B = I, \quad Q^B(z) = \mathbf{1}\{z < n\delta\}, \quad \lambda^B = 0.$$

Therefore the new aggregate productivity and the change in aggregate productivity are

$$\Phi_1(B) = n\delta, \quad \Delta\Phi \equiv \Phi_1(B) - \Phi_1(A) = \frac{n-1}{2}\delta.$$

4.3.1.1 FHK Decomposition

There is one continuer, no entry, and no within-firm productivity change. The FHK decomposition includes 5 terms: within (W), between (B), cross (C), entry (N), and exit (X). In this example, the terms are

$$W^{\text{FHK}} = 0, \quad B^{\text{FHK}} = \frac{(n-1)^2}{2n}\delta, \quad C^{\text{FHK}} = 0, \quad N^{\text{FHK}} = 0, \quad X^{\text{FHK}} = \frac{n-1}{2n}\delta.$$

Dividing by total growth $\Delta\Phi = \frac{n-1}{2}\delta$ gives the shares

$$\text{Within} = 0, \quad \text{Between} = 1 - \frac{1}{n}, \quad \text{Cross} = 0, \quad \text{Entry} = 0, \quad \text{Exit} = \frac{1}{n}.$$

Thus, even though this is a pure selection event (the only underlying policy change is that the regime switches from one with no exit to one with exit), FHK assigns only a $1/n$ share of total growth to exit and attributes the rest to between-firm reallocation. In the limiting case, as the number of firms approaches infinity, FHK will assign zero contribution to exit, despite the change in the underlying primitives suggesting that all of the change in aggregate productivity should be attributed to exit.

4.3.1.2 ($P, Q, \lambda G$) Decomposition

Since $P^B = P^A$ and there is no entry, our decomposition assigns the entire effect to exit:

$$\text{Within-all}^{OZ} = 0, \quad \text{Exit}^{OZ} = \frac{n-1}{2}\delta, \quad \text{Entry}^{OZ} = 0.$$

The corresponding shares are

$$\text{Within-all} = 0, \quad \text{Exit} = 1, \quad \text{Entry} = 0.$$

4.3.2 Example 2: Downward Drift, Threshold Exit, and Long Horizons

The purpose of this example is to show that standard accounting decompositions attribute exit effects based on firms' initial productivity and therefore miss the deterioration that occurs prior to exit, whereas our decomposition explicitly models the productivity process and correctly quantifies how exit can conceal future declines. In this example, each firm's productivity declines at a fixed rate over time, and firms exit upon declining to zero productivity. In this example, productivities are in log, so it is possible for them to become negative. Initial productivities are again

$$\delta, 2\delta, \dots, n\delta, \quad \Phi_0 = \frac{n+1}{2}\delta.$$

Under the post-reform regime,

$$P^B(z - \delta \mid z) = 1, \quad Q^B(z) = \mathbf{1}\{z \leq 0\}, \quad \lambda^B = 0.$$

Thus latent productivity declines deterministically by δ each period, and firms exit when the post-transition draw becomes nonpositive. Because exit occurs after the draw of next period's productivity, zero and negative productivity are never observed among active firms.

After h periods, the surviving firms have productivities

$$\delta, 2\delta, \dots, (n-h)\delta,$$

so the new aggregate productivity is

$$\Phi_h(B) = \frac{n-h+1}{2}\delta.$$

Hence total productivity growth relative to the pre-reform benchmark is

$$\Delta\Phi_h \equiv \Phi_h(B) - \Phi_h(A) = -\frac{h}{2}\delta.$$

4.3.2.1 FHK Decomposition

Over horizon h , the FHK terms are

$$W_h^{\text{FHK}} = -\frac{h(n-h)}{n}\delta, \quad B_h^{\text{FHK}} = \frac{h^2}{2n}\delta, \quad C_h^{\text{FHK}} = -\frac{h^2}{n}\delta$$

$$N_h^{\text{FHK}} = 0, \quad X_h^{\text{FHK}} = \frac{h(n-h)}{2n}\delta.$$

Dividing by total growth $\Delta\Phi_h = -\frac{h}{2}\delta$ gives the shares

$$\text{Within} = \frac{2(n-h)}{n}, \quad \text{Between} = -\frac{h}{n}, \quad \text{Cross} = \frac{2h}{n}, \quad \text{Entry} = 0, \quad \text{Exit} = -\frac{n-h}{n}.$$

Since total productivity growth is negative in this example, a positive level contribution corresponds to a negative share. The key point is that the exit contribution varies mechanically with the horizon. In this setting, exit protects the economy from further deterioration by removing firms whose productivity would otherwise continue to fall. However, FHK attributes exit effects based on firms' initial productivity rather than the productivity they would have had if they had remained active. As $h \rightarrow n$, many firms would have negative productivity absent exit, yet FHK cannot capture this protection from further decline and instead assigns nearly zero importance to exit.

4.3.2.2 $(P, Q, \lambda G)$ Decomposition

As before, to compute the contributions of each component of our decomposition, we use the Shapley value. In this example, there are two effective orderings: change P first and then Q , or change Q first and then P . We compute the marginal contribution of each component under both orderings and then average them.

Order 1: $P \rightarrow Q$.

First change P while holding $Q = Q^A$. The marginal contribution of within is

$$\Delta_P^{P \rightarrow Q}(h) = \Phi_h(P^B, Q^A) - \Phi_h(P^A, Q^A) = -h\delta.$$

Next change Q while holding $P = P^B$. The marginal contribution of exit is

$$\Delta_Q^{P \rightarrow Q}(h) = \Phi_h(P^B, Q^B) - \Phi_h(P^B, Q^A) = \frac{h}{2}\delta.$$

Order 2: $Q \rightarrow P$.

First change Q while holding $P = P^A$. Since productivities do not change under P^A , no firm reaches the exit threshold and therefore

$$\Delta_Q^{Q \rightarrow P}(h) = 0.$$

Next change P while holding $Q = Q^B$. The marginal contribution of within is

$$\Delta_P^{Q \rightarrow P}(h) = \Phi_h(P^B, Q^B) - \Phi_h(P^A, Q^B) = -\frac{h}{2}\delta.$$

Averaging across the two orderings gives

$$\text{Within-all}^{OZ}(h) = \frac{1}{2} \left(\Delta_P^{P \rightarrow Q}(h) + \Delta_P^{Q \rightarrow P}(h) \right) = -\frac{3h}{4}\delta,$$

and

$$\text{Exit}^{OZ}(h) = \frac{1}{2} \left(\Delta_Q^{P \rightarrow Q}(h) + \Delta_Q^{Q \rightarrow P}(h) \right) = \frac{h}{4}\delta.$$

Since entry is absent throughout, $\text{Entry}^{OZ}(h) = 0$.

Dividing by total growth $\Delta\Phi_h = -\frac{h}{2}\delta$ yields the shares

$$\text{Within-all} = \frac{3}{2}, \quad \text{Exit} = -\frac{1}{2}, \quad \text{Entry} = 0.$$

Unlike FHK, the contribution of exit is invariant to the horizon.

5 Identification and Estimation

This section describes how we recover the primitives of the productivity process – the transition matrix $P(z'|z)$, the exit rule $Q(z)$, and the entry process $\lambda G(z)$ – from firm-level

panel data. We first outline the observed data process, then state the identification result, and finally describe our estimation procedure. Throughout this section, we use tildes to denote observed empirical objects.

5.1 Observed Data Process

Firm-level data provide two directly observable objects: the empirical transition matrix $\tilde{P}(z'|z)$, which gives the probability that a firm observed with productivity z in period t is observed with productivity z' in period $t + 1$ conditional on survival, and the empirical exit rule $\tilde{Q}(z)$, which gives the probability that a firm with productivity z in period t exits in period $t + 1$. These differ from the underlying primitives because $P(z'|z)$ gives the transition rates for all firms, not just surviving firms, while $Q(z)$ gives the exit rule based on productivity in period $t + 1$, not productivity in period t . Under the timing assumption that firms first draw z' from $P(z'|z)$ and then exit with probability $Q(z')$, the relationship between the observed and structural objects is

$$\tilde{Q}(z) = \sum_{z'} P(z'|z)Q(z'), \quad \text{and} \quad (7)$$

$$\tilde{P}(z'|z) = \frac{P(z'|z)(1 - Q(z'))}{\sum_{z''} P(z''|z)(1 - Q(z''))}. \quad (8)$$

With our preferred timing, the observed entry distribution and mass of entrants equal the structural objects of interest. As such, we can use the directly observed empirical distribution of new firms, $\tilde{G}(z)$, and the empirical mass of entrants, $\tilde{\lambda}$, as estimators for the structural objects from the equalities $\tilde{G}(z) = G(z)$ and $\tilde{\lambda} = \lambda$.

5.2 Identification

Identification rests on the invertibility of the mapping from the latent (P, Q) to the observed $(\tilde{P}, \tilde{Q})$ (from Equation 7 and Equation 8). For a discretized model (so there are finitely many productivity states z), the following assumption is necessary and sufficient for identification.

Assumption 1. $\text{diag}(1 - \tilde{Q})\tilde{P}$ is full rank, where diag denotes the diagonalization operator.

Assumption 1 implies that $\text{diag}(1 - \tilde{Q})$ and $\tilde{P}$ are both individually full rank. Conceptually, this means (i) no productivity bin exits with certainty and (ii) observed transition patterns across bins are linearly independent (this prevents, for instance, two productivity bins from having identical observed transition rates).¹¹ This leads to the main identification result.

Proposition 1. *Suppose that Assumption 1 holds. Then the primitives are uniquely identified by*

- a. $Q = 1 - 1 \oslash \left(\text{diag}(1 - \tilde{Q})\tilde{P} \right)^{-1} 1$, where 1 denotes the vector of ones and $\oslash$ denotes element-wise division; and
- b. $P = \text{diag}(1 - \tilde{Q})\tilde{P}\text{diag}(1 - Q)^{-1}$.

Consistent and asymptotically normal estimators of P and Q follow directly from sample analogues of $\tilde{P}$ and $\tilde{Q}$. Proofs and asymptotic results are provided in Appendix F. We end the discussion of identification by remarking that Proposition 1 enables non-parametric identification of the processes governing the productivity dynamics in a large class of firm dynamics models (e.g. variations on [Hopenhayn \(1992\)](#)).

5.3 Estimation Procedure

While the identification result delivers a nonparametric estimator, this approach is numerically unstable in practice. We instead choose to estimate the parameters by constrained maximum likelihood. Let $X_i = (X_t^i, X_{t+1}^i)$ denote firm i 's observed productivity and exit status. The log-likelihood of the data given (P, Q) is

$$\mathcal{L}(X|P, Q) = \sum_{i=1}^n 1\{i \text{ exits}\} \log \tilde{Q}(X_t^i) + 1\{i \text{ survives}\} (\log P(X_{t+1}^i|X_t^i) + \log[1 - Q(X_{t+1}^i)]) . \quad (9)$$

¹¹As an example, point (i) precludes identification in models with a threshold exit rule, where firms below some productivity lower bound $\underline{z}$ exit with certainty.

We impose standard probability constraints: $0 \leq Q(z) \leq 1$, $0 \leq P(z'|z) \leq 1$ for all z and z' , and each row of P sums to one. We also impose an additional constraint to address an ill-conditioning problem that arises in this system. To motivate how this problem emerges, note that Equation 7 can be written in matrix form as $PQ = \tilde{Q}$. Suppose we treat $\tilde{Q}$ as observed and P as a primitive, so we can rearrange to obtain $Q = P^{-1}\tilde{Q}$. From this arrangement, it becomes clear that the condition number of P^{-1} (identical, by construction, to the condition number of P) measures how sensitive the estimate of Q is to deviations in $\tilde{Q}$: when P is ill-conditioned, small deviations in the observed $\tilde{Q}$ translate into large changes in the inferred Q . To mitigate this instability, we penalize the condition number of the estimated P , ensuring that small perturbations in observed data have minimal impact on the recovered estimate of Q .

To implement the penalty in practice, we modify the likelihood function by adding a regularization term on the condition number of P :

$$\max_{P,Q} \mathcal{L}(X|P,Q) - \kappa \cdot \text{cond}(P), \quad (10)$$

where $\kappa > 0$ controls the strength of the penalty. In Appendix F.1 we present details on implementation, tuning, and Monte Carlo experiments. The Monte Carlo evidence demonstrates that this regularization substantially improves the estimator's small-sample stability and accuracy.

As robustness, Appendix G reports derivations and Monte Carlo performance for a continuous, parametric AR(1)-style specification of the framework.

Having established how the primitives of the productivity process are estimated, the next section turns to the institutional context of China's state-sector reforms, which provides the empirical setting for our analysis.

6 Decomposing Productivity Growth in China’s State Sector

In this section, we apply our decomposition framework to study productivity growth among Chinese state-owned enterprises (SOEs) between 1998 and 2007. Our objective is to quantify the relative contributions of selective firm exit, within-firm productivity dynamics, and entry to the observed improvement in state-sector productivity. For brevity, throughout this section, references to “productivity growth” refer to productivity growth within the state sector unless explicitly noted otherwise.

We first estimate the underlying productivity process and document how its primitives evolve over the reform period. We then decompose productivity growth over the full transition and on a year-by-year basis: the full-transition decomposition shows that exit was the dominant driver over the reform period, while the year-by-year analysis reveals that productivity growth shifted from being driven primarily by exit in the early reform years to within-firm improvements later in the decade. We show that these findings are robust across alternative specifications. Finally, we show that standard accounting decompositions systematically understate the role of exit in this context by failing to account for the declining productivity trajectories of firms that leave the market.

6.1 Benchmark Estimation Specification

For our benchmark analysis, we consider the full transition from 1998–2007 as well as nine one-year intervals from 1998–1999 through 2006–2007. We measure firm-level productivity following [Hsieh and Klenow \(2009\)](#), using an elasticity of substitution of $\sigma = 3$ across firms within each industry. We normalize productivity within industry relative to the industry median and trim the bottom and top 1% of observations in each year. We discretize log productivity into ten bins, with bin edges chosen to balance uniform spacing and equal observation counts.¹²

¹²Given the choice of N_q bins with edges $z_2, \dots, z_{N_q}$ (with z_1 and z_{N_q+1} fixed as the minimum and maximum observed log TFP), defining the bin group indicator $q(z)$, and taking as given the weight on

We estimate the primitives of the productivity process – the transition matrix P , exit probabilities Q , and the entry distribution λG – using observed firm productivity and exit status. For the pre-reform regime, we assume the economy begins in a stationary distribution in 1998 with no exit or entry ($Q = 0$, $\lambda G = 0$), and infer P as the transition matrix consistent with stationarity. This assumption approximates the pre-reform SOE environment, in which firm exit was rare despite the formal existence of bankruptcy law.¹³

For the post-reform regime, we estimate P and Q using the nonparametric method described in Section 5, with penalty parameter $\kappa = 0.25$. We estimate the entry distribution λG directly from the empirical distribution of entrants, assuming no selection in entry. Together, these procedures recover all primitives required for the decomposition while remaining consistent with several key institutional features that underpinned China’s reform process.

6.2 Primitive Estimates

6.2.1 Transition Matrix Evolution

Before turning to the full decomposition, it is useful to examine what can be learned directly from the estimated primitives of the productivity process. In this subsection, we focus on the transition matrix for the full 9-year interval.¹⁴ To facilitate the analysis, we define simple measures of upward and downward mobility that can be extracted directly from the estimated transition matrices and that provide a transparent characterization of changes in productivity dynamics over the reform period.

We define upward mobility as the average number of productivity bins moved upward conditional on an upward transition, $\sum_{i>j}(i-j)P(j,i)$, and define downward mobility anal-

spacing, $\gamma \in [0, 1]$, and observations z_i for $i \in \{1, \dots, N\}$, the objective function is

$$\max_{\{z_j\}_{j=2}^{N_q}} \gamma \sum_{j=1}^{N_q} \log \left(\frac{z_{j+1} - z_j}{z_{N_q+1} - z_1} \right) + (1 - \gamma) \sum_{j=1}^{N_q} \log \left(\frac{1}{N} \sum_{i=1}^N 1_{\{q(z_i) = j\}} \right).$$

We pick $\gamma = 0.2$ in our benchmark estimation specification but test out other values to ensure γ does not affect the results.

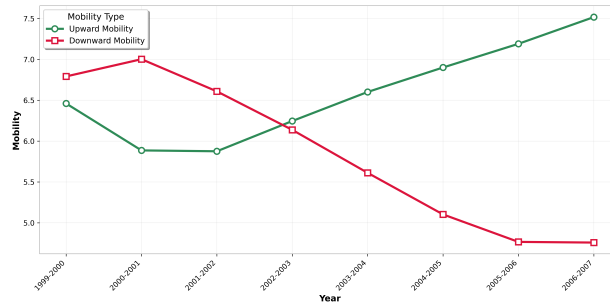
¹³The 1986 Enterprise Bankruptcy Law created the legal possibility of SOE bankruptcy; prior to this, there was no avenue for SOEs to go bankrupt. However, despite the passage of this law, [Hsieh and Song \(2015\)](#) documents an annual SOE exit rate of only 0.9% between 1991 and 1995.

¹⁴Estimated transition matrices for each one-year interval are presented in Appendix A.1, Figure 3.

ogously as $\sum_{i < j} (j - i)P(j, i)$. These measures summarize mobility patterns implied by the transition matrix, independent of exit.

As shown in Figure 4, upward mobility increases steadily over the reform period, rising from 6.5 in 1999 to approximately 7.5 in later years. At the same time, downward mobility declines from 6.7 in 1999 to be less than 5.0 by 2007. Together, these patterns indicate a shift in transition dynamics toward more frequent and larger upward movements and fewer downward movements.

Figure 4: Firms Productivity Mobility Trends: 1998 - 2007



Note: We interpolate the productivity mobility trend in 2004 as missing key variables in 2004 requires imputation, potentially affecting estimated transition kernels for 2003-2004 and 2004-2005.

6.2.2 Exit Probability Patterns

We now examine what can be learned directly from the estimated exit probabilities. Focusing on the relationship between productivity and exit allows us to characterize the degree of selection in the Chinese economy during the transition, as well as how it evolves over time, independent of the productivity transition process.

Estimated exit probabilities for each one-year interval are presented in Appendix A.1, Figure 4. In the early reform years (1998–2001), estimated exit probabilities are high across the productivity distribution. Low-productivity SOEs ($\log \text{TFP} < 5$) faced exit probabilities between 45–58%, while middle-productivity firms ($\log \text{TFP} \approx 5-8$) exhibited exit rates of roughly 30–40%. Even relatively productive firms faced substantial exit probabilities of 25–30%.

By the later reform period (2005–2007), exit probabilities declined markedly at all productivity levels: low-productivity firms faced exit rates of 32–35%, middle-productivity firms

experienced exit rates of 15–25%, and exit probabilities among high-productivity firms fell to approximately 10–15%.¹⁵ Of note is that the exit probabilities become less tilted over time, indicating weaker selection patterns.

6.3 Decomposition Results

Examining the estimated primitives in isolation shows that upward mobility increased, downward mobility declined, and exit probabilities fell over the reform period. The following two decompositions combine these primitives into a dynamic accounting of productivity growth by simulating how the productivity distribution would have evolved under alternative counterfactual sets of primitives.

We implement the decompositions as follows. Starting from the observed 1998 productivity distribution, we simulate the evolution of the distribution forward while varying the transition matrix P , exit probabilities Q , and the entrant distribution λG . Each component of the decomposition corresponds to the change in the distribution (or in weighted average productivity) generated by altering one primitive relative to a baseline. The specific implementation differs between the full-transition and year-by-year exercises, as detailed below.

We first present the cumulative decomposition over the full 1998–2007 transition and then turn to a year-by-year decomposition that highlights temporal patterns.

6.3.1 Decomposing the Reform: 1998–2007

For the full-transition decomposition, we follow the procedure described in Section 3. The contribution attributed to a component of the process is the effect of changing that component while holding all others fixed. Since the process is non-linear, changing the components in a different order will generate a different decomposition; as such, we follow the Shapley value approach and assign to each component its average contribution over all orderings. By construction, the components sum to the total 1998–2007 change.

¹⁵One persistent anomaly that appears across all years is the unexpected uptick in exit rates at very high productivity levels. This pattern may reflect measurement error in extreme productivity estimates, other data issues, or some other factors that fall outside our analysis. Despite this anomaly, the overall pattern of lower exit probabilities for more productive firms holds throughout the sample period.

An example ordering is the following: beginning from the 1998 distribution, we first allow the transition matrix P to change to its post-reform value while holding Q and λG fixed at pre-reform levels. We then allow Q to switch to its post-reform level, and finally allow λG to change. The change in the distribution at each step defines that channel's contribution.

Figure 5 presents the distributional decomposition.¹⁶ The black line in the figures represents the total change in the productivity density between 1998 and 2007: positive values indicate an increase in the mass of firms at a given productivity level, negative values indicate a decrease, and zeros indicates no change. The red bars represent the effect of changes in P , the green bars represent the additional effect of changes in Q , and the blue bars represent the additional effect of changes in λG .

Over the reform period, the productivity distribution shifts markedly to the right, with negative density changes at low productivity levels ($\log \text{TFP} \approx 2-7$) and positive changes at higher productivity levels ($\log \text{TFP} \approx 8-13$).

The exit effect dominates this transformation, contributing in the same direction as the overall change across the entire productivity distribution.¹⁷ This dominance reflects the strong and monotonic relationship between productivity and exit: recovered exit probabilities exceed 90% among the least productive firms and fall below 50% among the most productive firms (see Figure 5 in Appendix A.1). As a result, selective exit reinforces the observed shift in the distribution by disproportionately removing low-productivity firms.

The within-all effect exhibits a more heterogeneous pattern. It works against the overall change by increasing mass at the lower tail of the distribution ($\log \text{TFP} \approx 2-4$), while it reinforces the total change in the middle ($\log \text{TFP} \approx 4-8$) and upper tail ($\log \text{TFP} \approx 9-12$). This pattern indicates that within-firm changes contributed to dispersion, with high-productivity firms experiencing gains while low-productivity firms deteriorated.

The entry effect is negligible across most of the distribution and contributes positively only at the upper end ($\log \text{TFP} \approx 10-12$). This indicates that entrants were not systematic

¹⁶One important implementation detail is that while we have an annual panel, and could in theory compute a fixed post-reform annual process that utilizes all of the data, we instead use only the first and last years' data to compute this exercise. We treat parameters as jumping directly from the start of the reform to the end of the reform. This will be changed to the annual approach in a future draft.

¹⁷Since the pre-reform policy environment effectively barred SOEs from going bankrupt, the counterfactual policy we consider is one where firms that exit in the data would instead have been forced to continue operating through 2007.

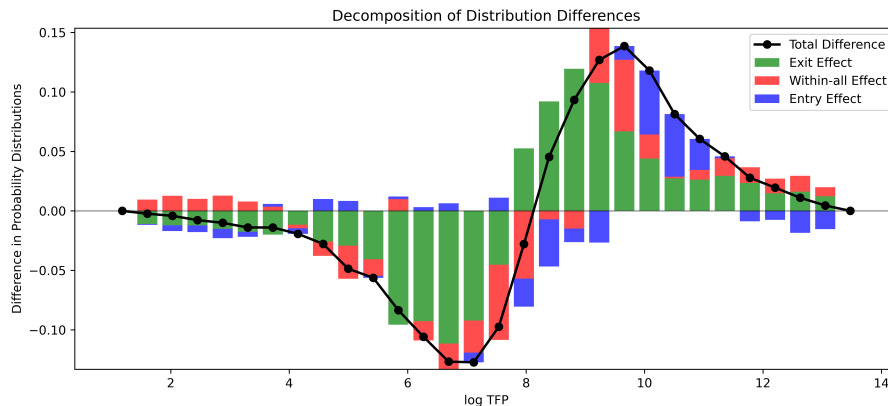
cally more productive than incumbents except at the highest productivity levels.

We summarize these distributional changes using the unweighted mean of log TFP. Mean log TFP increased by 1.25 log points between 1998 and 2007, with the exit effect accounting for 84.7% of this increase, the within-all effect accounting for 14.7%, and the entry effect contributing the remaining 0.06%.

As a robustness check, we compute the decomposition using input shares to construct a weighted mean of log TFP. Since input shares are not observed in 2007 for exiting firms, we impute them using 1998 values. Under this alternative measure, the exit effect contributes 85.8% of aggregate productivity growth, the within-all effect contributes -17.8% , and the entry effect contributes 32.1%. The larger contribution of entry under the weighted measure is consistent with evidence that entrants tend to be relatively large firms, amplifying their aggregate impact (see [Hsieh and Song \(2015\)](#)).

Taken together, the full-period decomposition shows that productivity growth during the reform period was driven primarily by selective exit, with secondary but non-trivial contributions from within-firm productivity dynamics and entry.

Figure 5: Benchmark Decomposition of Changes in TFP Distributions: 1998-2007



6.3.2 Decomposing the Reform: Year-by-Year

The full-transition decomposition attributes the cumulative change in the productivity distribution between 1998 and 2007 to sequential changes in the underlying primitives. We now consider a complementary exercise in which we simulate the distribution forward year-

by-year by allowing one primitive at a time to evolve from its pre-reform level while holding the others fixed. This approach allows us to trace the time-varying, cumulative impact of each channel in isolation, abstracting from interaction effects across primitives. Because each channel is evaluated separately, the resulting effects do not mechanically sum to the observed change in the weighted average productivity.¹⁸

Figure 6 reports the results. The top panel displays the cumulative change in mean log TFP attributable to each primitive over time relative to the 1998 baseline, while the bottom panel shows yearly marginal effects.

The year-by-year decomposition reveals two distinct phases of transformation. During the early reform period (1998–2002), productivity gains were driven primarily by exit. The exit channel contributed 0.172, 0.126, and 0.115 log points in the first three years, before declining steadily to less than 0.1 log point annually by the end of the reform period. In contrast, the within-all effect was initially negative (approximately -0.07 log points) but turned positive after 2001 and increased sharply thereafter. By 2006–2007, within-firm dynamics contributed 0.277 log points, compared to 0.035 log points from exit. Entry played a stable but relatively modest role throughout the transition period, contributing between 0.03 and 0.06 log points annually.

These timing patterns align with the evolution of the estimated primitives documented in Section 6.2. Over the reform period, the declining contribution of exit reflects weaker selection, as exit probabilities become less steeply decreasing in productivity, while the rising contribution of the within-all channel mirrors the surge in upward mobility documented in the transition matrices.

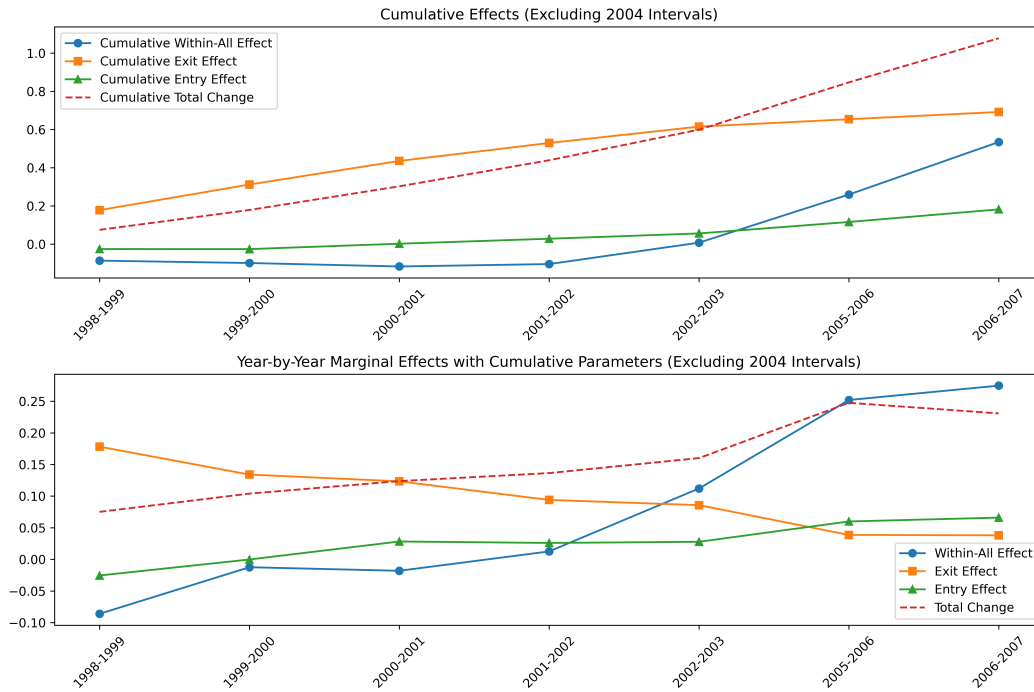
Note that because each annual effect isolates a single changing primitive while holding others fixed, the sum of the individual channels in a given year need not equal the observed year-to-year change in mean productivity. The difference reflects interactions among exit, within-firm dynamics, and entry when all channels operate simultaneously. However, this does not affect our interpretation of the temporal patterns recovered from the year-by-year

¹⁸In Section 6.3.1, by using the Shapley value approach described in Section 4, the sum of the contributions of each component equals the total change in productivity, i.e. there are no residual interaction terms. In this section, on the other hand, we do not follow the Shapley value approach. We instead model the effects of each component independently. This approach permits interaction terms, thus the sum of the contributions from each component is not guaranteed to sum to the total change in productivity.

decomposition exercise.

As a robustness check, we repeat the exercise using the input-share weighted mean of log TFP (see Appendix A.1 Figure 6). The qualitative pattern is unchanged: exit effects, initially the primary driver of productivity growth, decline in importance over time, while the within-firm channel, which contributes negatively to growth early in the transition, becomes the dominant contributor after 2005.

Figure 6: Temporal Pattern of Effects in Driving Average TFP Changes during 1998-2007



Note: Missing key variables in 2004 required imputation, potentially affecting productivity measures for 2004-2005. We exclude the 2004 decomposition results in this figure.

6.4 Robustness Checks

We conduct an extensive set of robustness checks by varying key aspects of the estimation procedure. These include alternative penalty parameters, different bin configurations (varying both the number of bins and the weighting between equal spacing and equal observation counts), and adjustments to sample trimming thresholds. We also consider alternative productivity measures beyond the [Hsieh and Klenow \(2009\)](#) specification and perform industry – specific estimation to account for sectoral heterogeneity. Finally, we replicate the

entire estimation and decomposition procedure using a continuous, parametric AR(1)-style specification. Across all exercises, the central findings – the dominant role of exit in the early reform period and the rising importance of within-firm dynamics thereafter – remain unchanged (see Appendix G for details).

6.5 Decomposing the Reform: Existing Accounting Decompositions

In this subsection, we present results from applying several standard accounting decompositions to our data, then explain why these methods can be misleading in the presence of strong selection over long horizons.

Table 1 reports aggregate productivity decompositions using several standard accounting methods. Panel A presents results for the full reform period, while Panels B and C repeat the exercise over shorter horizons. By construction, all methods reproduce the same aggregate productivity change; differences arise only in how this change is attributed to within-firm improvements, reallocation among surviving firms, entry, and exit.

We begin with exit. The estimated contribution of exit is highly sensitive to the time horizon over which the decomposition is computed. Over short intervals, especially in the early reform years, exit appears quantitatively important. For example, in the 1999–2001 period, exit accounts for 51 percent of productivity growth in the [Foster et al. \(2001\)](#) (FHK) decomposition and 66 percent in the [Melitz and Polanec \(2015\)](#) (DOPD) decomposition. In later two-year windows, the estimated contribution of exit remains positive but becomes smaller, consistent with the descriptive evidence in Section 3 that selection was strongest early in the reform period. However, as the interval length increases, the contribution attributed to exit declines sharply. The four-year decompositions produce smaller exit estimates than the two-year decompositions, and the full-period decompositions produce smaller estimates than the four-year decompositions. Indeed, for both FHK and DOPD, the exit contribution over the full reform period lies below the range of the corresponding short-horizon estimates. Thus, the longer the decomposition horizon, the closer the estimated role of exit moves toward zero. In the next subsection, we describe what drives this pattern and why it motivates us

to develop an alternative decomposition framework.

Entry exhibits a related sensitivity, though for a somewhat different reason. In the FHK and [Griliches and Regev \(1995\)](#) (GR) decompositions, the contribution of entrants depends on how entrant productivity is compared to historical benchmark productivity levels. FHK compares entrants to the previous period's aggregate productivity, while GR compares them to an average benchmark across the two periods. As a result, the estimated entry contribution varies substantially across interval lengths and can take on implausibly large values. In contrast, the DOPD decomposition compares entrants to contemporaneous incumbents and produces more stable results across horizons. In our data, this method assigns negative entry contributions in most periods, indicating that entrants are on average less productive than incumbents. This pattern is consistent with the descriptive evidence in Section 3, where entrants appear less productive than surviving firms in both the early and late reform years.

The extreme entry and exit values in [Baily et al. \(1992\)](#) (BHC) arise because, unlike the other decompositions, it does not benchmark entrants and exiters against a reference productivity level. This makes the two terms mechanically large and difficult to interpret.

The within-firm component is much more stable across decompositions. Although magnitudes differ, the methods broadly agree on the time pattern: within-firm productivity growth is weak, and in some cases negative, early in the reform period, but becomes much more important later. Over the full reform period, the within component is consistently large. This pattern is again consistent with the empirical evidence in Section 3, which shows substantial productivity growth among firms that survive from 1998 to 2007. Relative to exit and entry, then, the within component appears considerably more robust across accounting methods.

Finally, the between or reallocation component is positive in nearly all specifications, indicating that productivity growth was accompanied by a shift in market share toward more productive surviving firms. The magnitude of this component varies across methods, sometimes substantially, but the sign is generally stable. The contribution of reallocation also tends to be larger earlier in the transition and smaller in later periods, suggesting that much of the reallocation occurred during the initial phase of reform. Over the full reform period, however, the between component remains positive, pointing to an overall contribution

of improved allocative efficiency among survivors.

Taken together, the standard accounting decompositions deliver a mixed picture. On the one hand, they consistently indicate positive within-firm productivity growth over the full reform period and positive reallocation among surviving firms. On the other hand, the contributions of exit and entry are highly sensitive to the decomposition horizon and to the choice of benchmark productivity levels.

6.5.1 Why Standard Accounting Decompositions Mislead in Our Setting

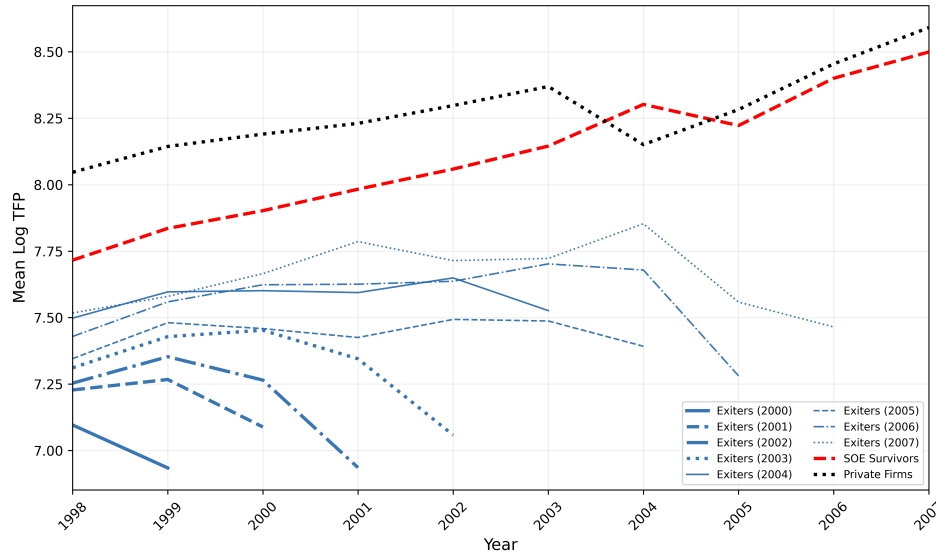
The horizon example from Section 4.3.2 provides a simple explanation for the anomalies in Table 1. Standard accounting decompositions evaluate exit using firms' productivity at the beginning of the interval. Over short horizons, the set of exiters is concentrated among the least productive firms, so exit appears to contribute strongly to aggregate productivity growth. Over longer horizons, however, the set of eventual exiters expands to include firms that were more productive initially but only later declined and exited. As a result, the average initial productivity of exiters rises mechanically with the horizon, which drives down the exit contribution in standard accounting decompositions even if selective exit remains economically important throughout.

Table 1: Aggregate TFP Decomposition: Comparison Across Accounting Methods

Period	$\Delta\Phi$	Method	Within	Between	Cross (% of $\Delta\Phi$)	Entry	Exit
<i>Panel A: Full Periods</i>							
1998–2007	1.569	BHC	33.0	51.8	–	264.1	-248.9
		GR	38.9	9.6	–	13.1	38.4
		FHK [†]	33.0	6.8	11.8	35.3	13.0
		Dyn. OP	53.0	36.9	–	-16.4	26.5
1999–2007	1.447	BHC	32.5	68.6	–	266.1	-267.2
		GR	39.6	10.9	–	12.3	37.2
		FHK [†]	32.5	7.8	14.3	32.8	12.5
		Dyn. OP	54.5	34.9	–	-14.1	24.7
<i>Panel B: Four-Year Periods</i>							
1999–2003	0.712	BHC	36.0	103.5	–	327.8	-367.3
		GR	40.5	17.0	–	6.1	36.4
		FHK [†]	36.0	16.0	8.8	19.5	19.6
		Dyn. OP	34.0	46.4	–	-9.9	29.5
2003–2007	0.681	BHC	45.9	99.1	–	299.3	-344.3
		GR	49.8	23.6	–	-4.0	30.7
		FHK [†]	45.9	22.3	7.8	7.1	16.9
		Dyn. OP	60.4	35.8	–	-19.5	23.3
<i>Panel C: Two-Year Periods</i>							
1999–2001	0.181	BHC	-12.2	263.5	–	795.2	-946.5
		GR	-0.9	55.3	–	-16.8	62.5
		FHK [†]	-12.2	46.1	22.6	-8.0	51.4
		Dyn. OP	26.3	39.0	–	-31.2	66.0
2001–2003	0.508	BHC	56.7	65.4	–	243.0	-265.2
		GR	53.1	14.0	–	0.8	32.1
		FHK [†]	56.7	19.2	-7.2	7.9	23.3
		Dyn. OP	24.0	55.1	–	-7.3	28.3
2003–2005	0.269	BHC	49.1	333.7	–	468.2	-751.0
		GR	48.3	43.2	–	-29.4	37.9
		FHK [†]	49.1	48.4	-1.6	-22.0	26.2
		Dyn. OP	6.2	102.9	–	-43.3	34.2
2005–2007	0.385	BHC	78.0	-97.0	–	397.8	-278.8
		GR	75.7	10.1	–	-22.0	36.1
		FHK [†]	78.0	10.3	-4.6	-13.4	29.7
		Dyn. OP	92.9	9.9	–	-36.9	34.1

Notes: $\Delta\Phi$ is the change in log aggregate TFP (identical across methods by construction). All component columns report shares of $\Delta\Phi$ in percentage points; shares sum to 100 within each method. **BHC**: [Bailey et al. \(1992\)](#). **GR**: [Griliches and Regev \(1995\)](#); within uses $\bar{s}_i = (s_{i1} + s_{i2})/2$, between uses $\bar{z}_i - \bar{\Phi}$. **FHK[†]**: [Foster et al. \(2001\)](#) with explicit cross term (between uses $z_{i1} - \Phi_1$). **Dyn. OP**: Dynamic Olley–Pakes ([Melitz and Polanec \(2015\)](#)); Within = $\Delta\bar{z}_S$ (change in unweighted survivor mean); Between = Δcov_S (change in allocative efficiency among survivors). Input-share weights throughout.

Figure 7: Productivity Trajectories of Exiting State-Owned Firms: 1998 - 2007



Note: We track firms with more than one observation to capture pre-exit productivity trends. Year of exit is defined as the first year a firm is absent from the dataset.

Figure 7 shows direct evidence of this mechanism in the data. As the data horizon is extended from 1998 to 2007, the average productivity in 1998 of firms that exit by the terminal year rises steadily. In other words, the longer the horizon, the less negatively selected the set of eventual exiters appears when measured using only their initial productivity. At the same time, Figure 7 shows that these firms are persistently less productive than survivors and typically experience additional declines prior to exit. Together, these patterns help explain why existing accounting decompositions assign a shrinking role to exit at long horizons: they evaluate exiters based on where they started, not on the deterioration that occurs before they leave the market.

7 Interpretation through the Lens of Hopenhayn

China's state-owned enterprise (SOE) reforms between 1998 and 2007 generated one of the most remarkable periods of state-sector productivity growth ever observed. Two central questions are what mechanisms drove this growth and what role each mechanism played.

Two broad explanations have been proposed for the first question.

The first explanation emphasizes improvements in managerial incentives within incumbent firms. Reforms during this period introduced corporatization, performance contracts, and stronger monitoring of SOE managers, potentially improving effort, technology adoption, and organizational efficiency. Under this view, productivity growth reflects improvements in the productivity dynamics of surviving firms (e.g., [Groves et al., 1994](#); [Jefferson et al., 1996](#); [Hsieh and Song, 2015](#)).

The second explanation emphasizes the hardening of budget constraints and the resulting selection of firms. Prior to reform, many low-productivity SOEs remained in operation despite persistent losses because of government support. Exit among SOEs was extremely rare: [Hsieh and Song \(2015\)](#) report that the annual exit rate of state-owned firms was only about 0.9 percent between 1991 and 1995. The restructuring wave associated with the policy of “grasp the large, let go of the small” therefore potentially raised aggregate productivity by removing persistently unproductive firms (e.g., [Brandt et al., 2012](#); [Gu and Jia, 2022](#)).

Both mechanisms can generate substantial improvements in aggregate productivity, but they operate through fundamentally different margins: improvements in productivity dynamics within firms versus stronger selection through firm exit.

Distinguishing between these mechanisms requires going beyond our statistical primitives and toward a framework that links institutional reforms to the forces governing firm dynamics. One candidate is to use the Hopenhayn-style model discussed in Section 4.1.4, which can formalize the link between policy forces and observed dynamics. Then, our statistical decompositions can be used as suggestive evidence that observed changes in productivity dynamics were plausibly driven by particular policy reforms. The insights gained from such a model should be understood as suggestive rather than definitive. Our framework does not by itself identify the deep institutional sources of changes in P , Q , or λG . Rather, the point of this discussion is to demonstrate that a Hopenhayn-style model provides a transparent benchmark linking alternative reform mechanisms to different objects in our statistical decomposition. Interpreted through that benchmark, our results can be used to assess which explanations are more consistent with the decomposition results presented in Section 6, without claiming to fully identify the underlying policy sequence.

Through the lens of this model of firm dynamics, management improvements are naturally represented as changes in the productivity process itself. Better management, improved incentives, or more effective restructuring would directly shift the law of motion for firm productivity, and therefore appear as changes in the transition object P . Such changes could also generate partial equilibrium responses in exit, since more productive firms are less likely to fail, and general equilibrium responses in both exit and entry, since a more productive incumbent sector may affect equilibrium prices. By contrast, hardening budget constraints operates through the continuation decision. This most closely corresponds to changes in the fixed cost of operating through a reduction in government subsidies. Such changes have a direct effect on the exit rule Q , and may also affect entry through equilibrium prices, but do not alter the underlying productivity transition process P , since it is treated as exogenous.

This distinction provides a useful benchmark for interpreting our estimates. If the main effect of reform operated through harder budget constraints, one would expect the reform to appear primarily through changes in exit behavior, with little or no change in the all-firm productivity transition process. If instead the main effect operated through improvements in management or firm organization, one would expect to see changes in the productivity transition process itself, potentially accompanied by second order changes in exit and entry. Through this lens, our decomposition results are consistent with a particular sequencing.

Our decomposition results show that in the early years of reform (1998–2002), productivity growth is driven mainly by exit, consistent with a first stage in which harder budget constraints intensified selection. This timing aligns with the introduction of the “grasp the large, let go of the small” policy that encouraged exit among small, unproductive SOEs through harder budget constraints and stricter bankruptcy enforcement, while also encouraging consolidation among larger and more productive SOEs.

In later years (2003–2007), however, our decomposition shows that productivity growth is driven mainly by changes in the within-all-firms transition process, which is difficult to explain with budget hardening alone and is more naturally interpreted as evidence of improvements in managerial or organizational performance. This timing aligns with the establishment of the State-Owned Assets Supervision and Administration Commission (SASAC)

in 2003 and the passage of the split-share structure reform in 2005, both of which were implemented with the goal of improving corporate governance.

This sequencing reconciles the literature by showing that both stories play prominent roles at different times, and aligns closely with the timing and objectives of the various reforms implemented during this period.

8 Conclusion

This paper studies the role of selection in the productivity growth of Chinese state-owned enterprises during the 1998–2007 reform period. We show that standard accounting decompositions face two important shortcomings in this setting. First, as selection becomes stronger, these decompositions tend to reassign the effect of exit to between-firm reallocation, obscuring the role of market selection. Second, over long horizons they mechanically drive the contribution of exit toward zero, even when selective exit is central to aggregate productivity growth. To address these issues, we develop a new decomposition framework that separately identifies within-all-firms productivity transitions, exit, and entry, allowing us to trace how each margin contributes to the evolution of the productivity distribution. Because these objects map naturally into the primitives of Hopenhayn-style models of firm dynamics, the framework is both flexible and economically interpretable. Applied to the Chinese SOE reform, it yields a new interpretation of the reform period: early productivity growth appears to have been driven primarily by stronger selection through exit, while later growth appears to have been driven primarily by improvements in the productivity transition process. Read through the lens of a Hopenhayn-style model, this pattern is most consistent with an early phase in which hardening budget constraints intensified selection, followed by a later phase in which changes in managerial incentives, governance, or firm organization improved firms' underlying productivity dynamics. This reconciles previous conflicting views in the literature. More broadly, the framework provides a simple way to connect decomposition results back to plausible policy mechanisms.

For future work, we propose extending our selection model to address the double-selection problem faced by state-owned firms: selection into privatization and selection to remain in

the market. [Gu and Jia \(2022\)](#) take an important step in this direction by modeling the endogenous choice between exit, privatization, and corporatization, and finding that the transformation decision is strongly related to firm productivity and size. Integrating this structural transformation margin into our diagnostic decomposition framework would make it possible to assess whether accounting for both forms of selection changes conclusions about the relative importance of exit versus ownership transformation in driving aggregate productivity gains. A second extension is to broaden the outcome space beyond productivity alone. Incorporating measures such as size, profitability, innovation output, and return on capital could provide a more complete picture of how institutional change affects the joint dynamics of firm behavior and performance. Together, these extensions would deepen our understanding of the reform period and shed further light on the dynamics of state capitalism.

More generally, the framework can be applied in settings beyond the Chinese SOE reform. One natural direction is to allow the state space to be multidimensional, so that firm dynamics are characterized jointly by productivity and other firm attributes, and to allow for multiple margins of selection rather than exit alone. Another is to incorporate measurement error. Finally, the framework also suggests a new approach to the calibration of Hopenhayn-style models: rather than imposing a parametric law of motion and inferring it indirectly from selected moments, researchers can use panel data to recover the key dynamic objects nonparametrically and then impose additional structure only where economic interpretation requires it.

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